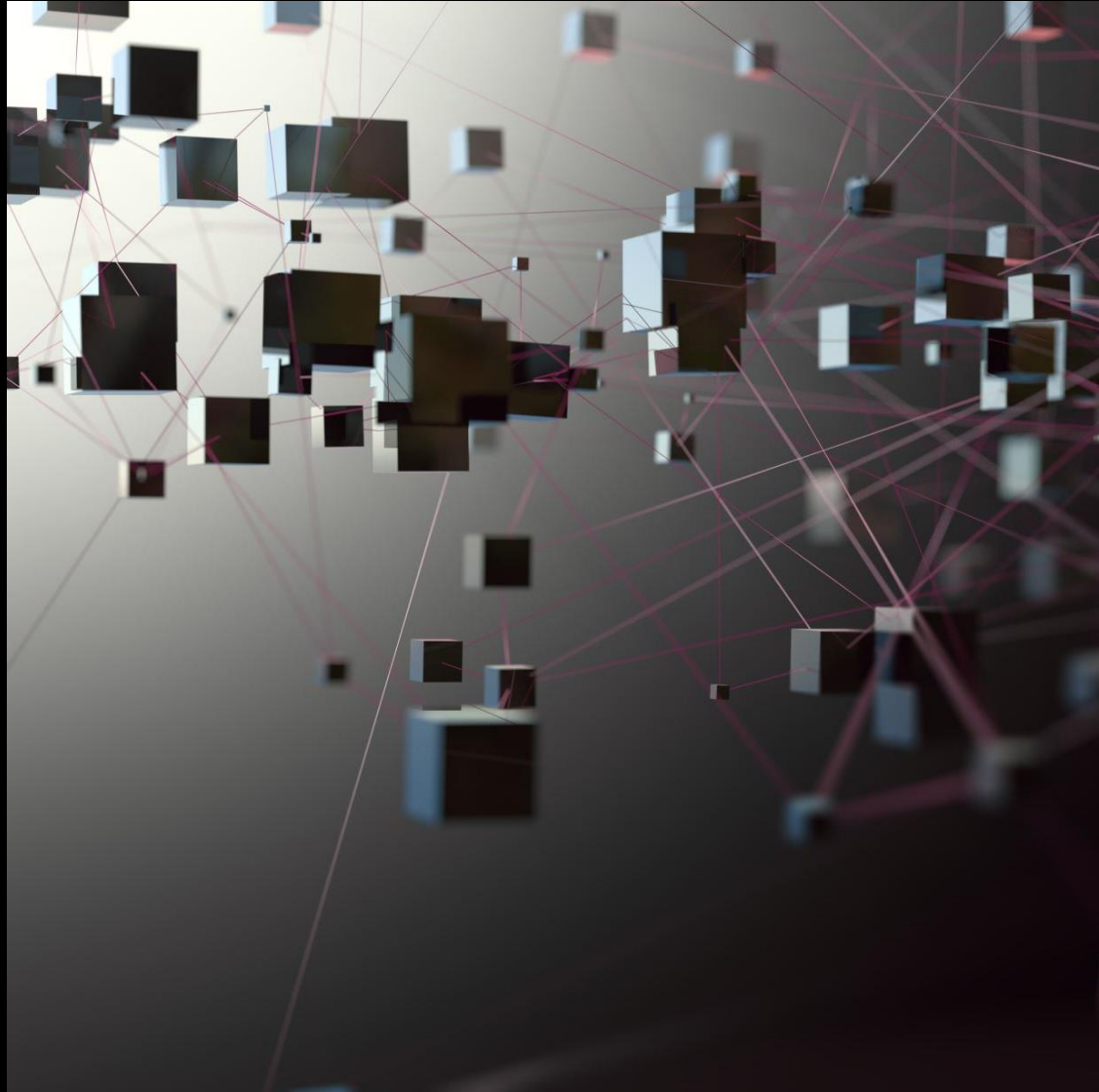
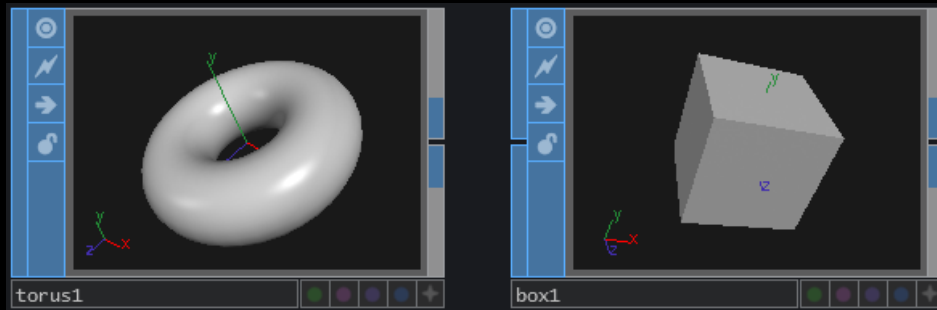


TOUCHDESIGNER 3D-GEOMETRY



THE MAIN OPERATORS

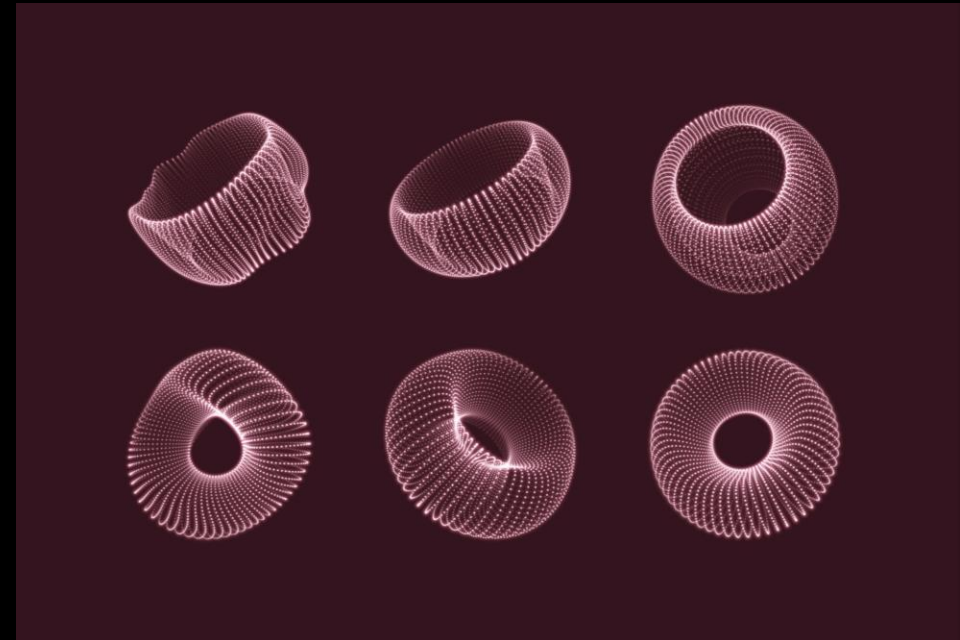


SOP – SURFACE OPERATORS

- Generate
- Import
- Manipulate 3D-surfaces or also called geometry

MAIN SUPPORTED SURFACE TYPES

-
- 3D points
 - Polygons
 - Lines
 - Surfaces
 - Spheres
 - meatballs



A DIFFERENCE IN SYSTEMS



- In earlier versions of Touchdesigner *PARTICLES* and *POINTCLOUDS* were also made in SOP's. It showed that working on larger projects it became too CPU straining and they moved them to the TOP domain resulting in GPU rendering and a big relieve for your computer.

A FIRST LOOK AT ALL THE SOP'S

OP Create Dialog

Search

COMP

TOP

CHOP

SOP

MAT

DAT

Custom

Add	Deform	Material	Select
Alembic	Delete	Merge	Sequence Blend
Align	Divide	Metaball	Skin
Arm	Extrude	Model	Sort
Attribute	Face Track	Noise	Sphere
Attribute Create	Facet	Null	Spring
Basis	File In	Object Merge	Sprinkle
Blend	Fillet	Oculus Rift	Sprite
Bone Group	Fit	OpenVR	Stitch
Boolean	Force	Out	Subdivide
Box	Fractal	Particle	Superquad
Bridge	Grid	Point	Surfsect
Cache	Group	Polyloft	Sweep
Cap	Hole	Polypatch	Switch
Capture	Import Select	Polyreduce	Text
Capture Region	In	Polyspline	Texture
Carve	Inverse Curve	Polystitch	Torus
CHOP to	Iso Surface	Primitive	Trace
Circle	Join	Profile	Trail
Clay	Joint	Project	Transform
Clip	Kinect	Rails	Trim
Convert	Lattice	Raster	Tristrip
Copy	Limit	Ray	Tube
CPlusPlus	Line	Rectangle	Twist
Creep	Line Thick	Refine	Vertex
Curveclay	LOD	Resample	Wireframe
Curvesect	LSystem	Revolve	ZED
DAT to	Magnet	Script	

OP Create Dialog

?

x

Search

COMP

TOP

CHOP

SOP

MAT

DAT

Custom

Add	Deform	Material	Select
Alembic	Delete	Merge	Sequence Blend
Align	Divide	Metaball	Skin
Arm	Extrude	Model	Sort
Attribute	Face Track	Noise	Sphere
Attribute Create	Facet	Null	Spring
Basis	File In	Object Merge	Sprinkle
Blend	Fillet	Oculus Rift	Sprite
Bone Group	Fit	OpenVR	Stitch
Boolean	Force	Out	Subdivide
Box	Fractal	Particle	Superquad
Bridge	Grid	Point	Surfsect
Cache	Group	Polyloft	Sweep
Cap	Hole	Polypatch	Switch
Capture	Import Select	Polyreduce	Text
Capture Region	In	Polyspline	Texture
Carve	Inverse Curve	Polystitch	Torus
CHOP to	Iso Surface	Primitive	Trace
Circle	Join	Profile	Trail
Clay	Joint	Project	Transform
Clip	Kinect	Rails	Trim
Convert	Lattice	Raster	Tristrip
Copy	Limit	Ray	Tube
CPlusPlus	Line	Rectangle	Twist
Creep	Line Thick	Refine	Vertex
Curveclay	LOD	Resample	Wireframe
Curvesect	LSystem	Revolve	ZED
DAT to	Magnet	Script	

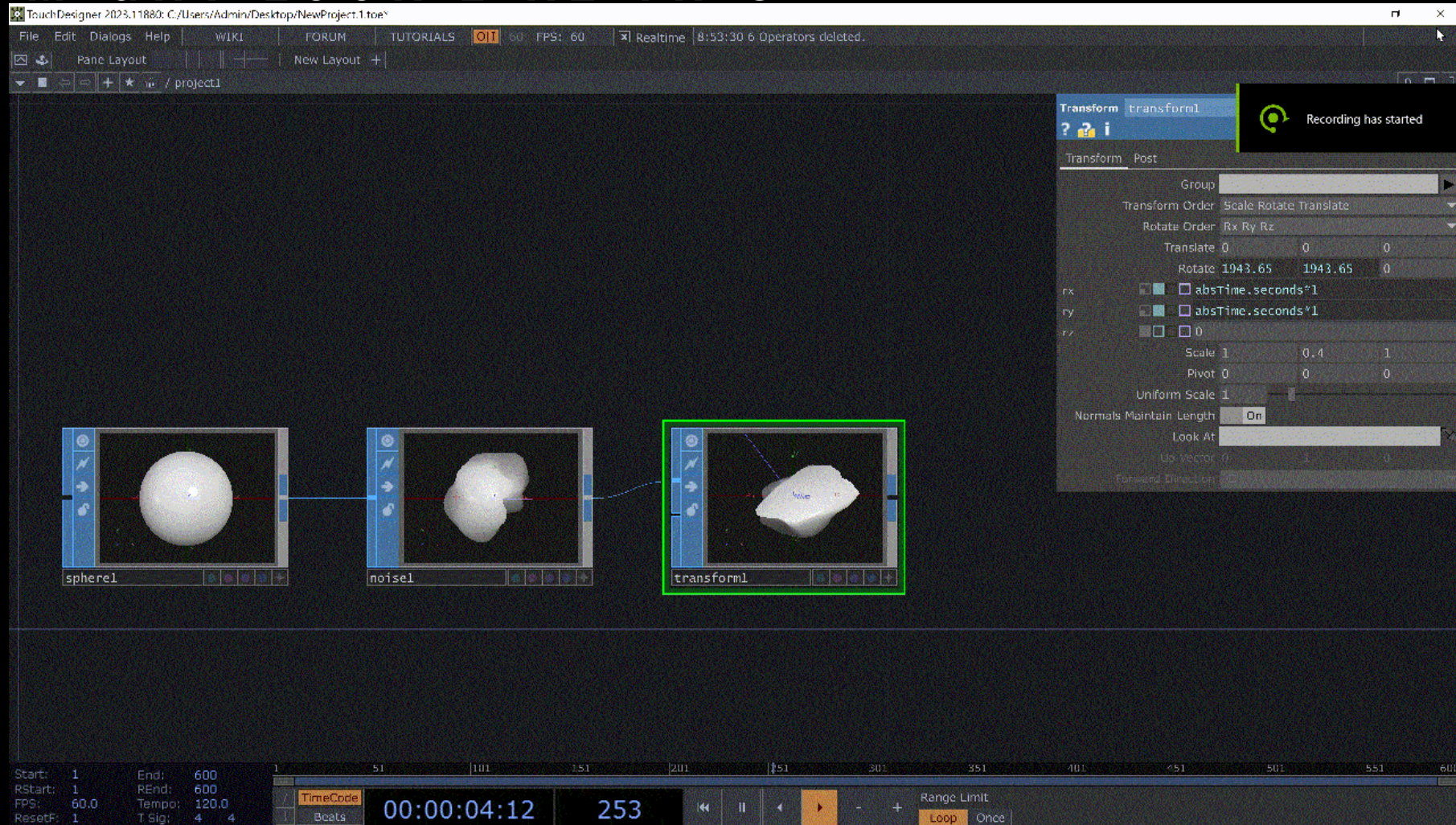
Couple of operators might look familiar to last week for example:

- Noise
- File in
- Switch

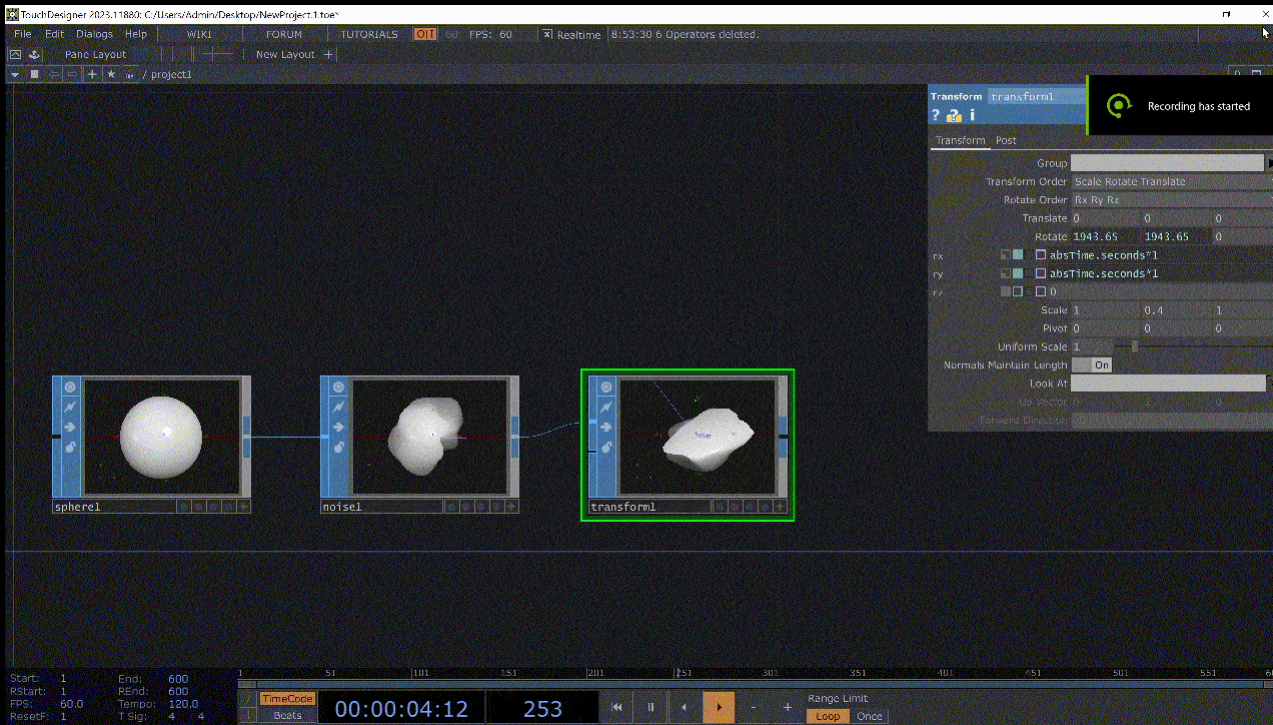
But probably most of them are new. Basic geometry shapes are:

- BOX
- CIRCLE
- METABALL
- RECTANGLE
- SPHERE
- TORUS
- TUBE

IF WE CREATE A RANDOM SOP NETWORK IT MIGHT LOOK LIKE THIS



OBSTACLE!



- Unfortunately SOP stays in the surface domain and cannot be mixed on the fly into video or TOP domain.
- Hence I needed to make this crappy GIF
- So we need some way to convert the image, not the data, into the TOP domain.

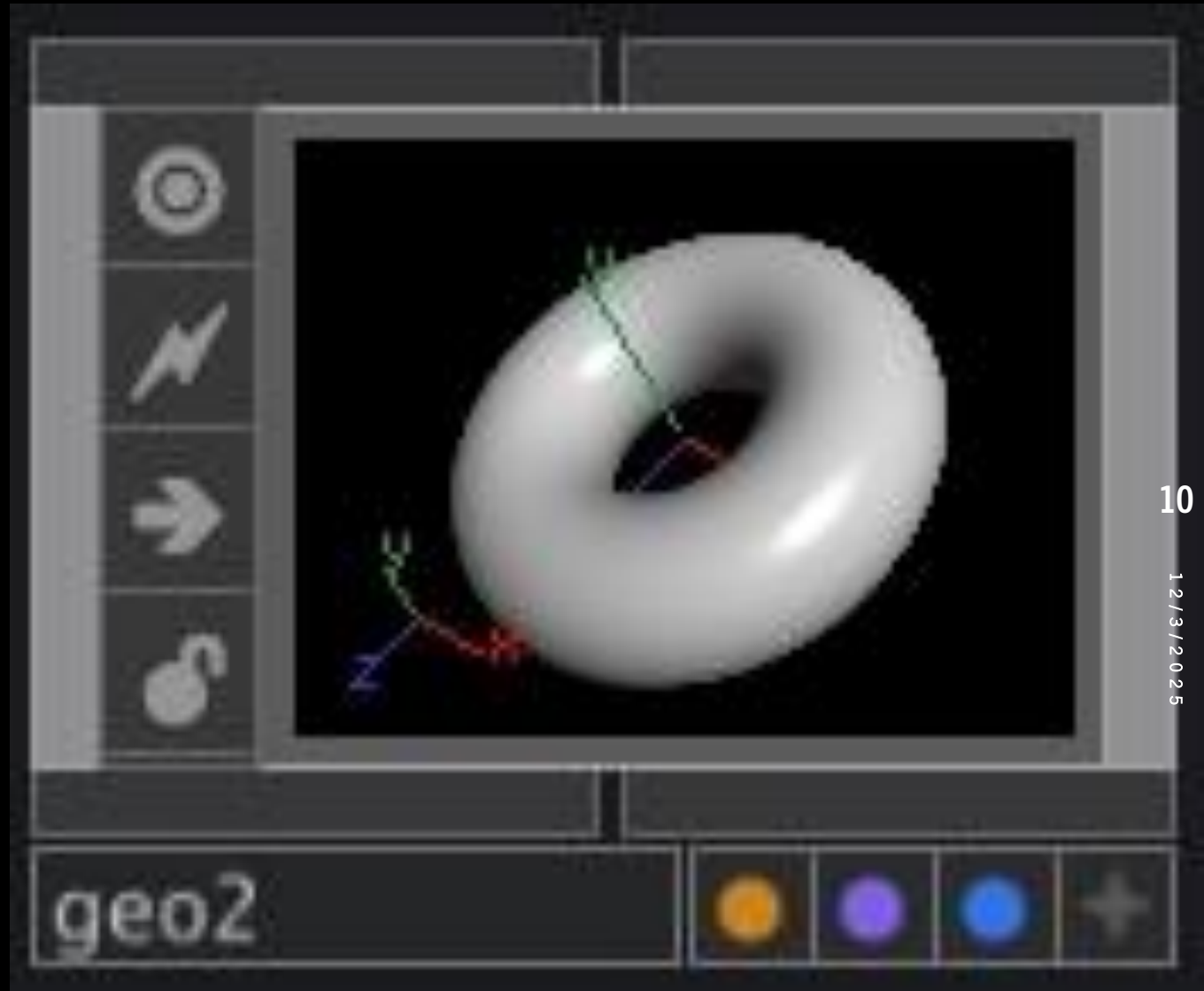
INTRODUCING: GEOMETRY COMP



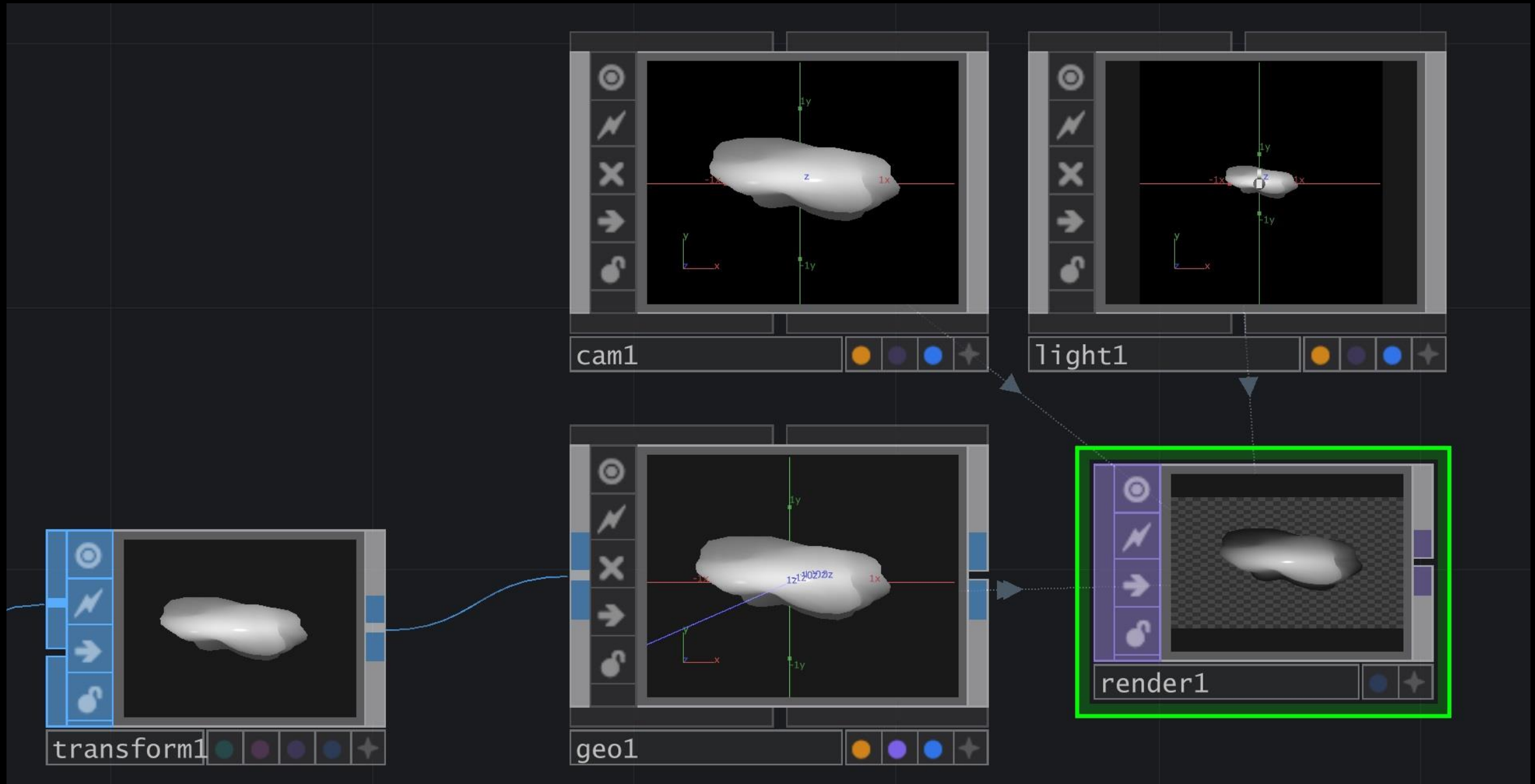
The GEOMETRY COMP can be used at the *end* of your SOP network to convert it into TOP picture information. It needs a couple of additional operators in order to do so successfully.

- Camera COMP
- Light COMP
- Render TOP
- Geometry COMP

These four operators live in the heart of your SOP and TOP combined networks.



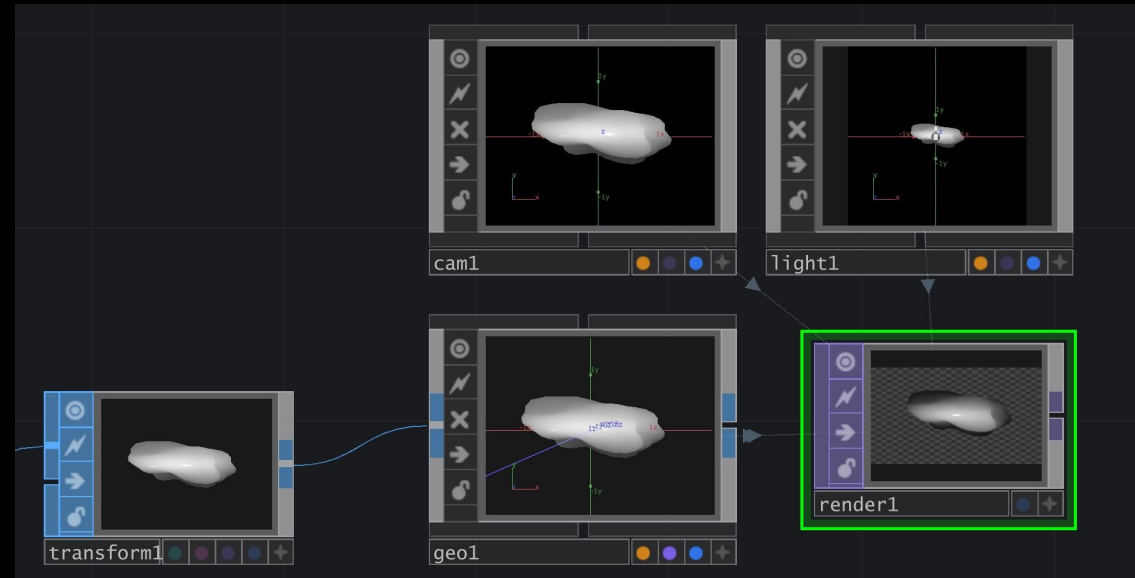
A SUCCESSFUL CONVERSION FROM SOP TO TOP



A COUPLE OF CONFUSING THINGS

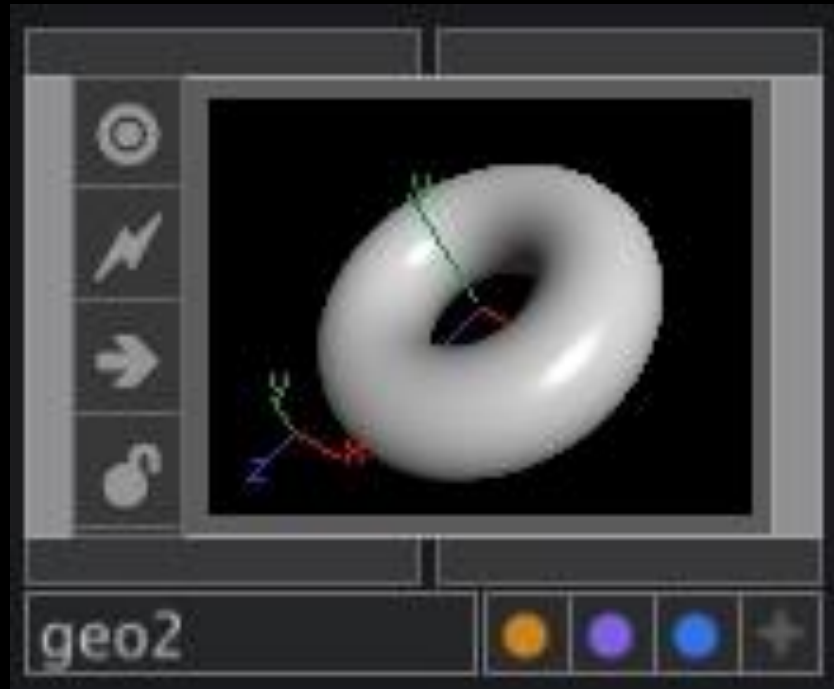
When going from SOP to TOP through a geometry COMP, camera COMP and a light COMP to a render some confusing things might occur to you.

- The geometry COMP changes appearance
- The geometry COMP is still blue, referencing SOP but goes to TOP?
- There are no wires between geometry, camera, light COMP and render TOP?



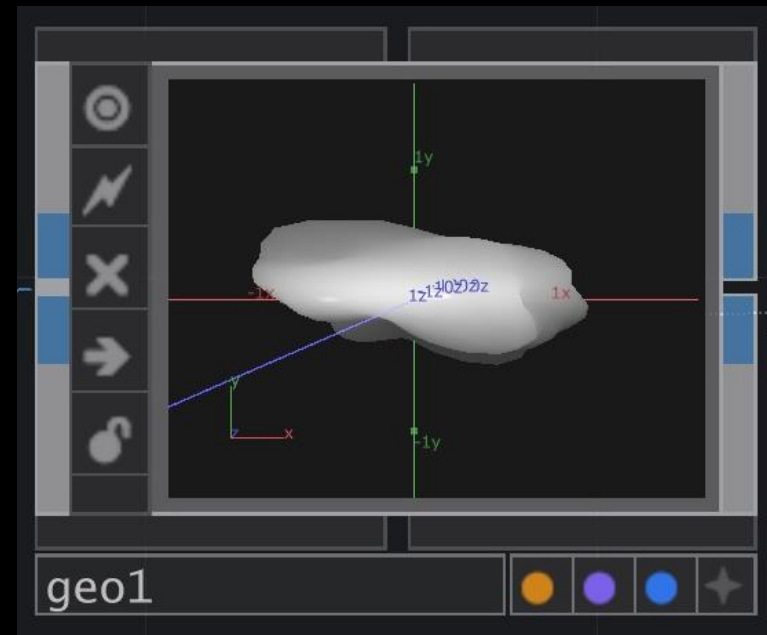
CHANGE OF APPEARANCE

- NO INPUT
- NO OUTPUT
- NO ASSOCIATED OP FAMILY



WHEN CONNECTED TO A SOP

- INPUT
- OUTPUT
- ASSOCIATED WITH SOP FAMILY



HOW TO CONNECT THE LAST SOP TO THE GEOMETRY COMP WHEN THERE ARE NO INPUTS?

YOU RIGHT CLICK ON THE OUTPUT OF THE LAST SOP IN YOUR NETWORK.
RIGHT CLICK WILL AUTO OPEN OP
CREATE DIALOG > COMP > GEOMETRY



YOU FIRST LEFT CLICK ON THE
OUTPUT SO YOU SEE THE WIRE > TAB
ON THE KEYBOARD > COMP >
GEOMETRY

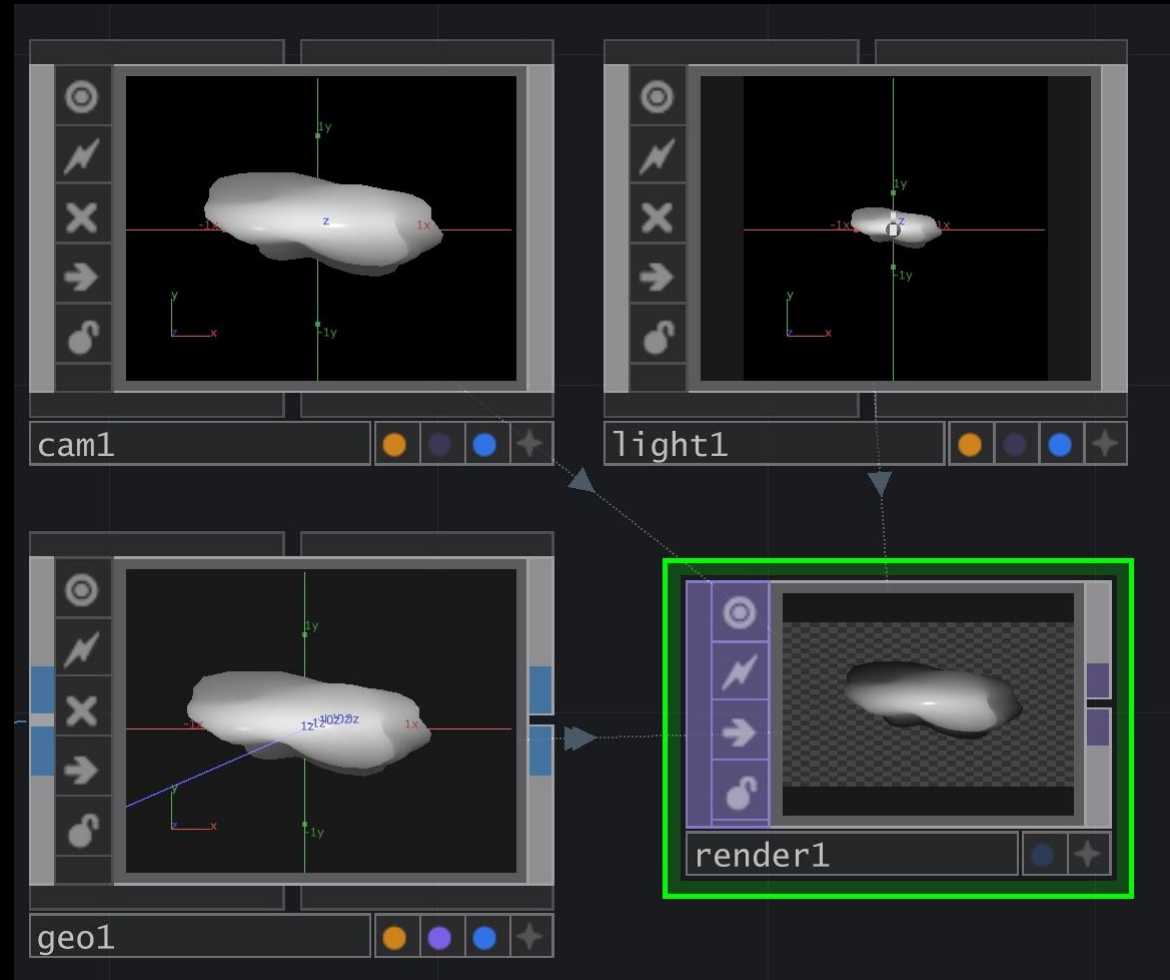


WHERE DID THE WIRES GO?

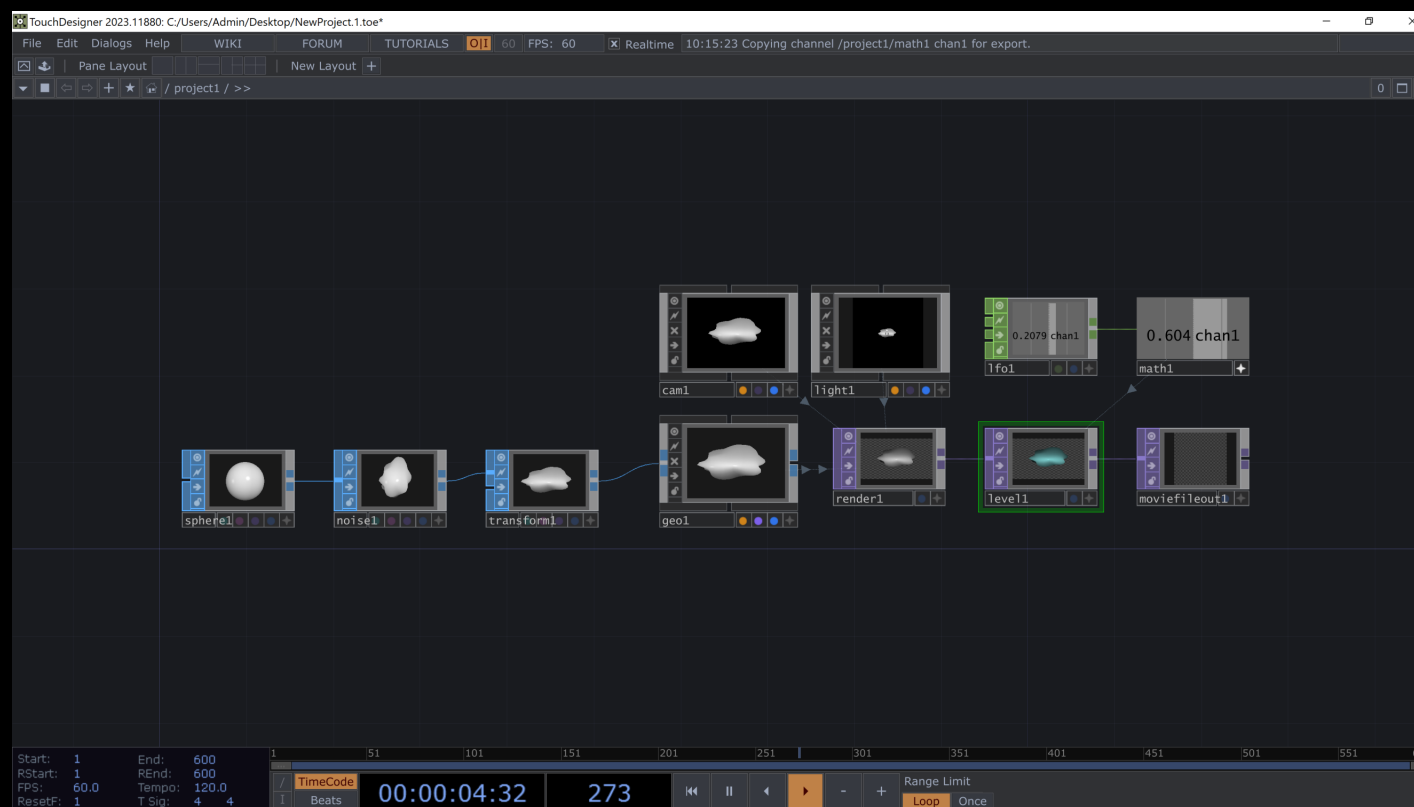
There are multiple connections between operators possible in Touchdesigner. The two most common are:

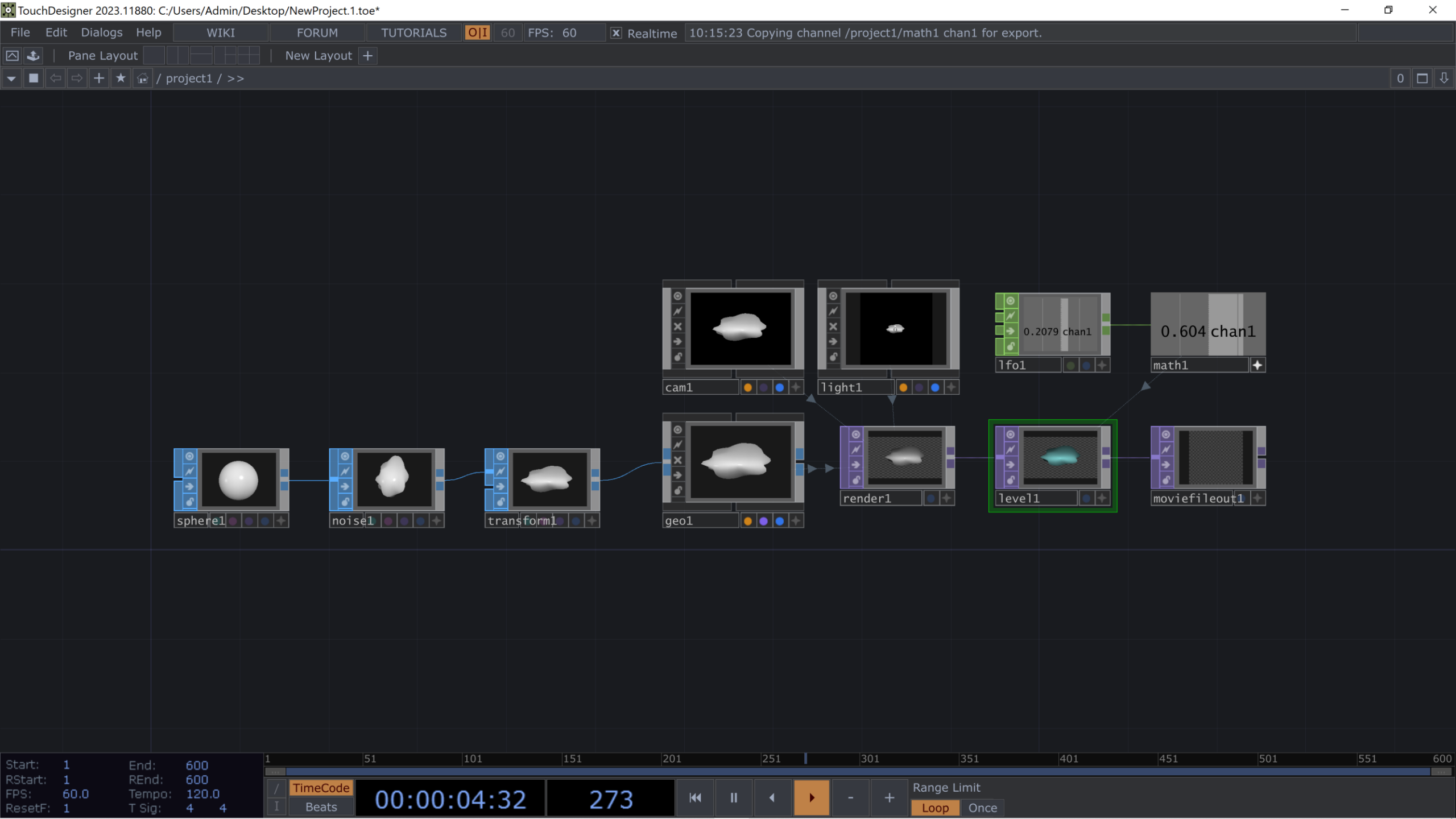
- Wires
- Dotted lines

We know the wires and the dotted lines we also saw last week. A dotted line means that information is being referenced between different operators. Within or mixing families. The small arrow traveling over the dotted line indicated the flow of information so you know where it is coming from and where it goes to.

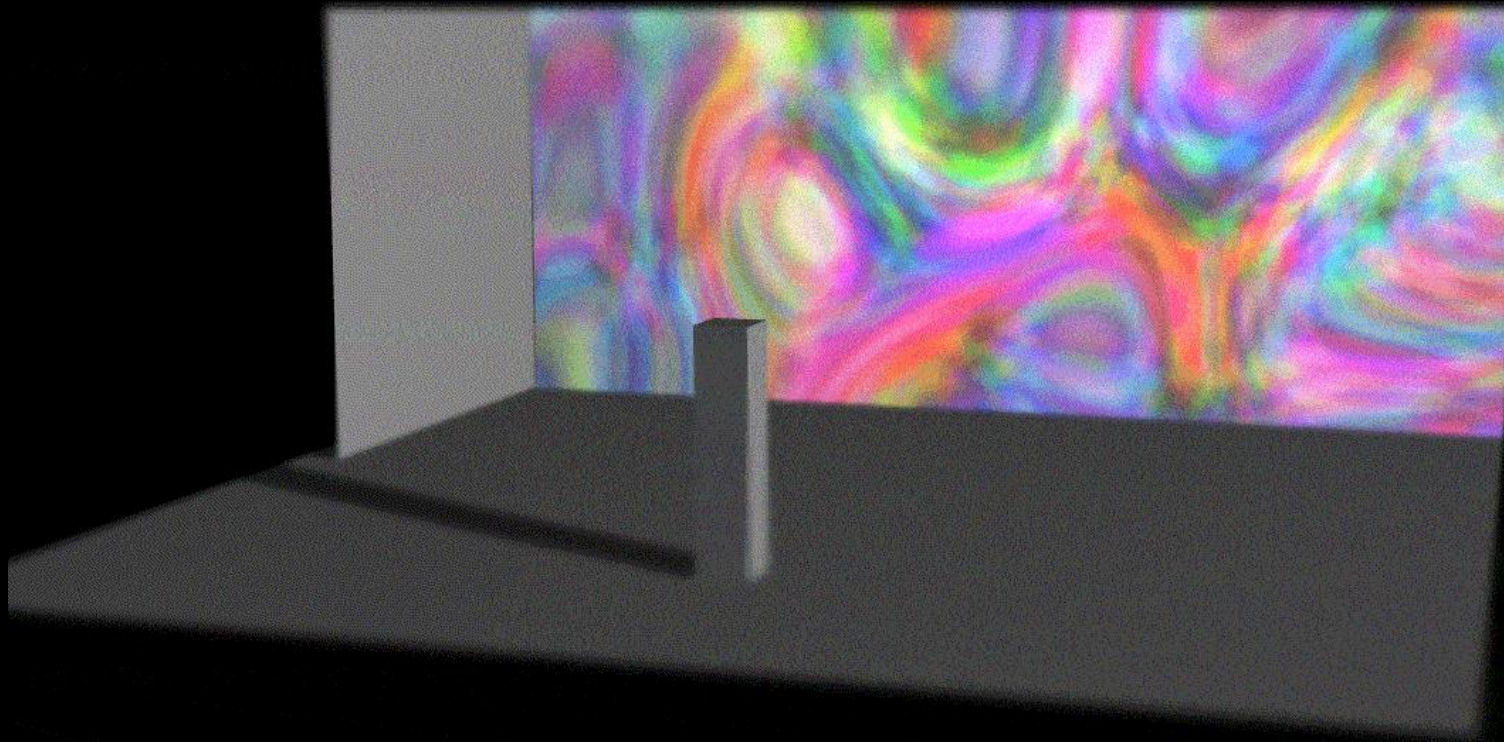


AFTER RENDER TOP YOU CAN BUILD YOUR NETWORK
AS WE PRACTICED LAST WEEK

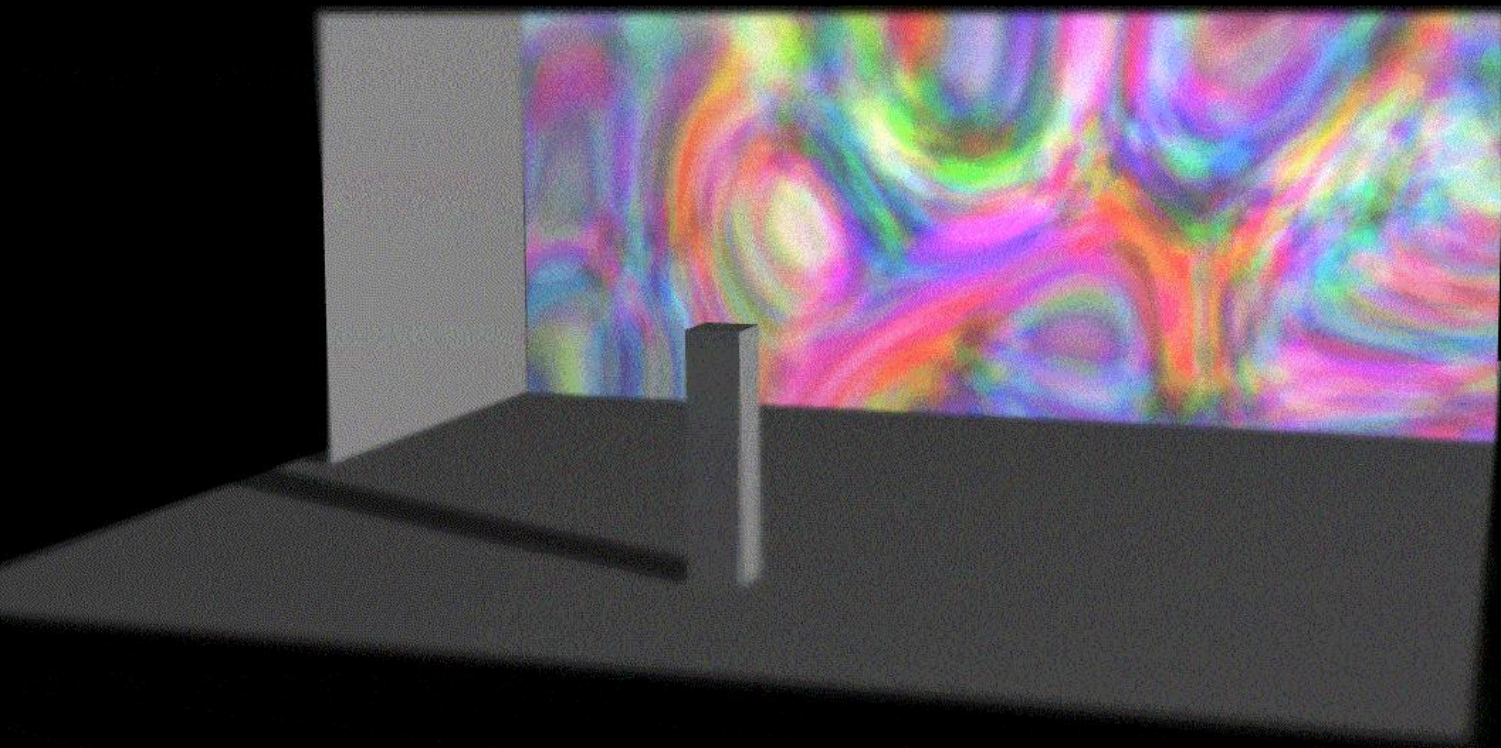




CASE: A SITUATIONAL RENDER



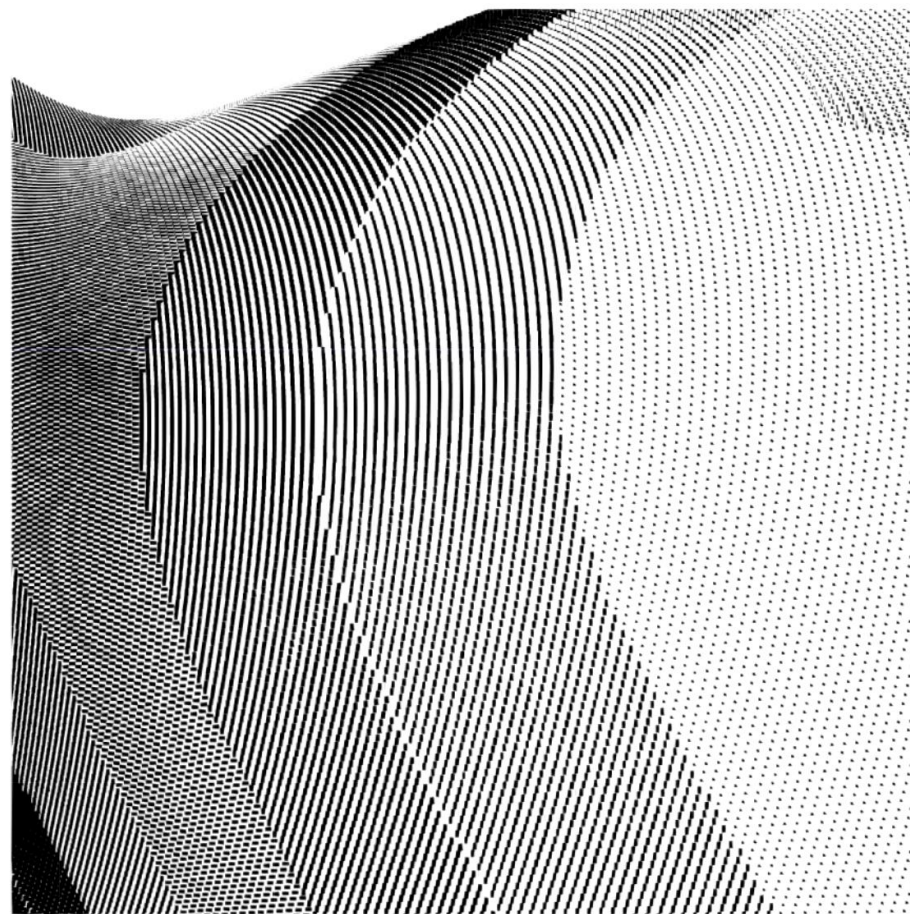
CASE: A SITUATIONAL RENDER BREAKDOWN



- Multiple surfaces
- Geometry comp for each surface
- One camera
- Two lights, one as projector one for normal lighting
- Some form of animation
- Render top
- Tilt-shift effect

INSTANCING

We're going to build this
minimalistic generative poster. As
you can see it can be used as a still
but the original file is animated.



Starling Moire

INSTANCING

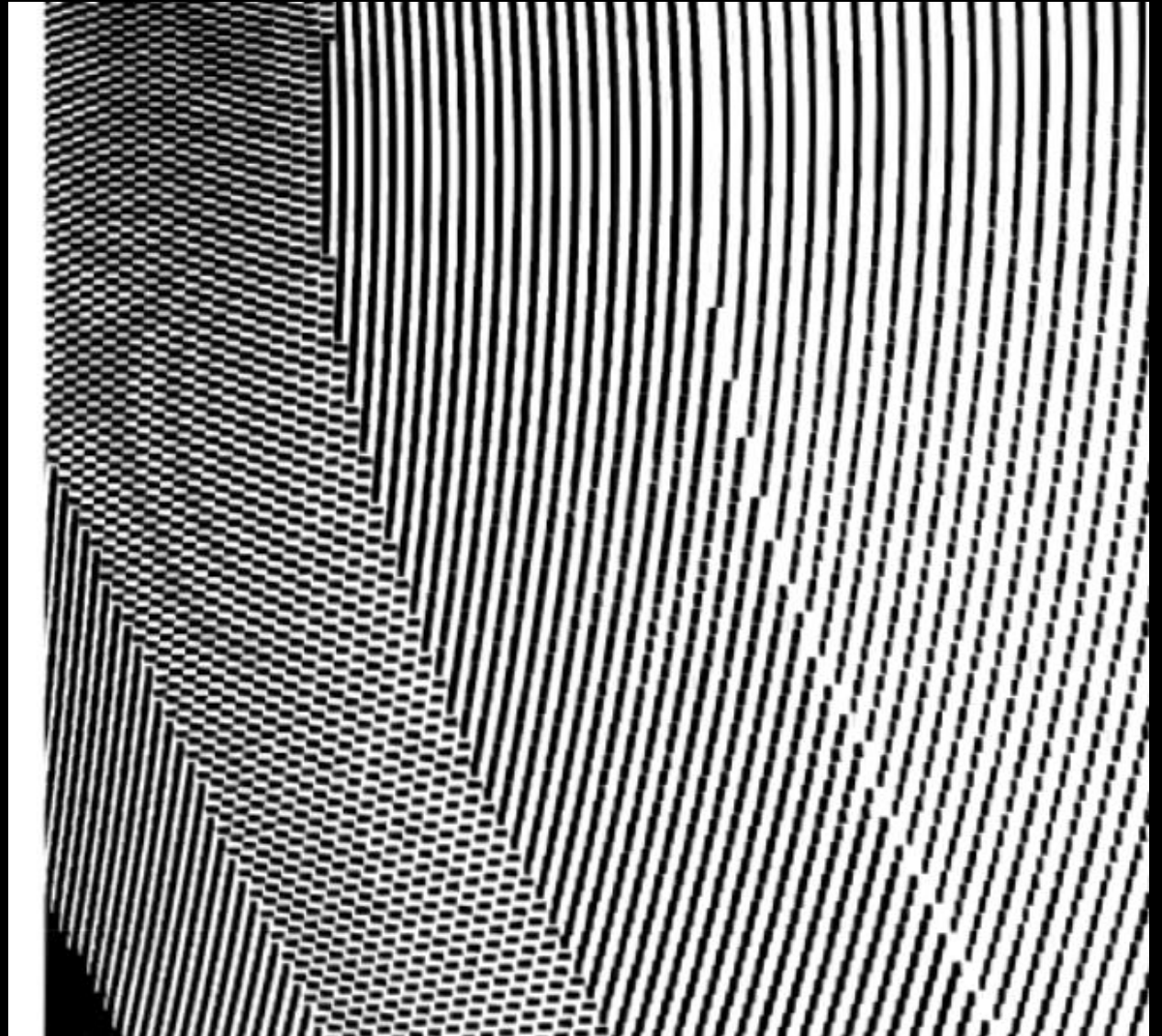
Do you see all the individual elements that build up the starling-like flock of the poster?

They are all individual pieces of geometry with materials applied to them. If we would have to copy all these pieces of geometry and render them inside a SOP environment our computer will choke on the amount of calculations.

Why is that?

In Touchdesigner's inherent architecture SOP or geometry is calculated through the CPU. The CPU in your computer is a linear device, meaning it does calculations line by line from top to bottom.

However, video information is calculated using the GPU in your computer. The GPU or Graphics Processing Unit can do parallel calculations meaning that it can do many many calculations at once.



INSTANCING

To overcome the problem of too many geometry elements in your project and your computer choking out on them we have a technique called instancing available to us.

Instancing is a way of making instances, or referenced copies, of an element. The copying or instancing of this piece of geometry is done through the geometry comp operator. To reduce calculative resources usually we use low level information to do the instancing.

We could use pixel color information to do the instancing. If we create two ramps of different solid colors and opposite directions we have a X and Y axis ready for us.

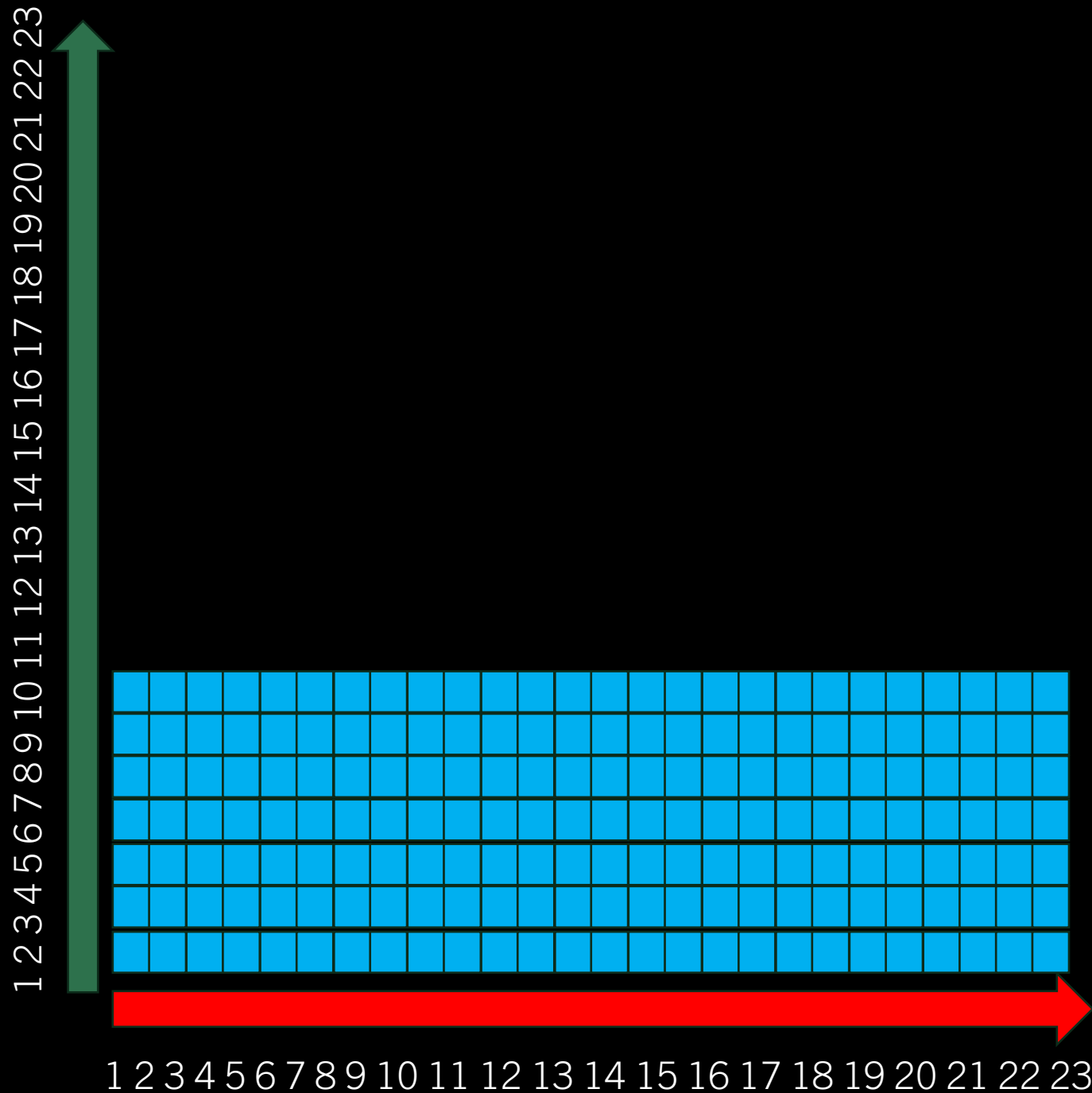
X axis = red ramp = black pixel = index 0; full red pixel = index 255

Y axis = green ramp = black pixel = index 0; full green pixel = index 255

Important with instancing is that the amount of instances have to be the same on all axes or ways of instancing.

Therefore referencing resolutions to different sources of instancing is advised so when you change resolution on one the others are also updated.





This graph shows in a minimal simplistic way how instancing works. Let's say the blue rectangle is the piece of geometry that should be instanced.

We use a green ramp for the Y-axis and a red ramp for the X-axis. For every new pixel point in the ramp, in this case going from 1 to 23, an instance of the rectangle will be placed.

Now we're working with a referenced coordinate system. If the coordinates are static the grid is also static. But it does not have to be static we can manipulate the coordinates to generate movement and displacement within the instancing.

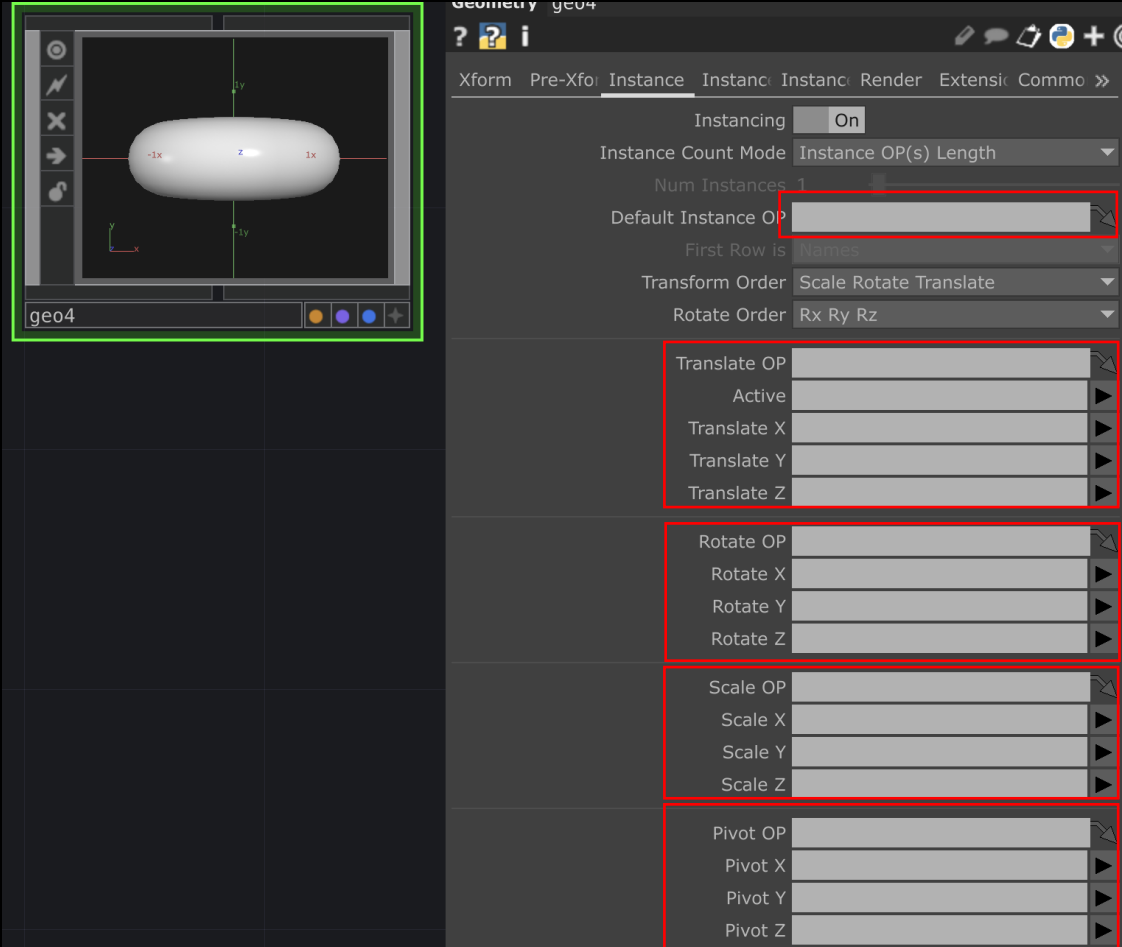
WHERE TO FIND INSTANCING

Instancing can be accessed through the Geometry COMP operator. It has its own tabs and has to be manually activated.

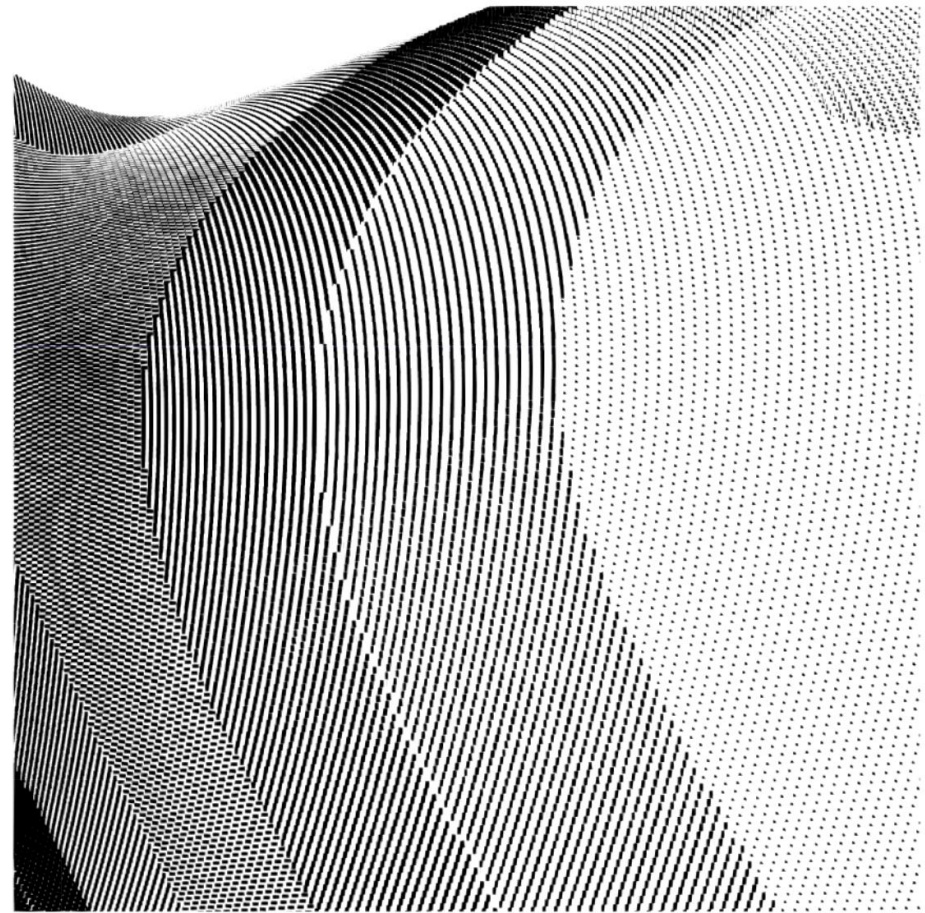
The red boxes on the image besides we can see there are a lot of different operations on how to apply instances. We can use it for translation, rotation, scaling or pivoting. However we can also instance textures, colors and other parameters.

You can use a default instance operator that will do all the instancing on different operations. Or you can assign different operators to do different instancing tasks. However the uniformal amount if instances still applies.

The line of thought behind this is: say you have 550 instances of translation on the X and Y-axis generating a grid. If you want to apply instanced color to that grid of 600 instances the amount mismatches and therefore cannot be assigned resulting in an error.



CASE:
GENERATIVE
INSTANCED
POSTER



Starling Moire

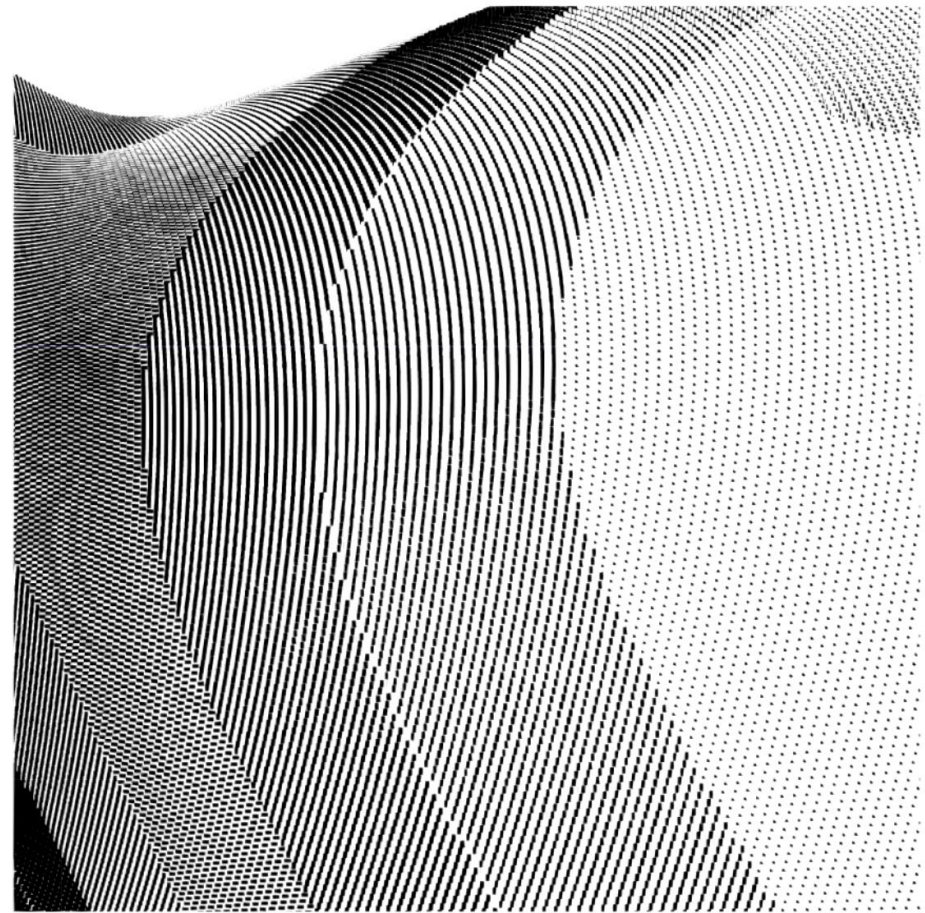
CASE: GENERATIVE INSTANCED POSTER

Color information for instancing

Displacement of instancing

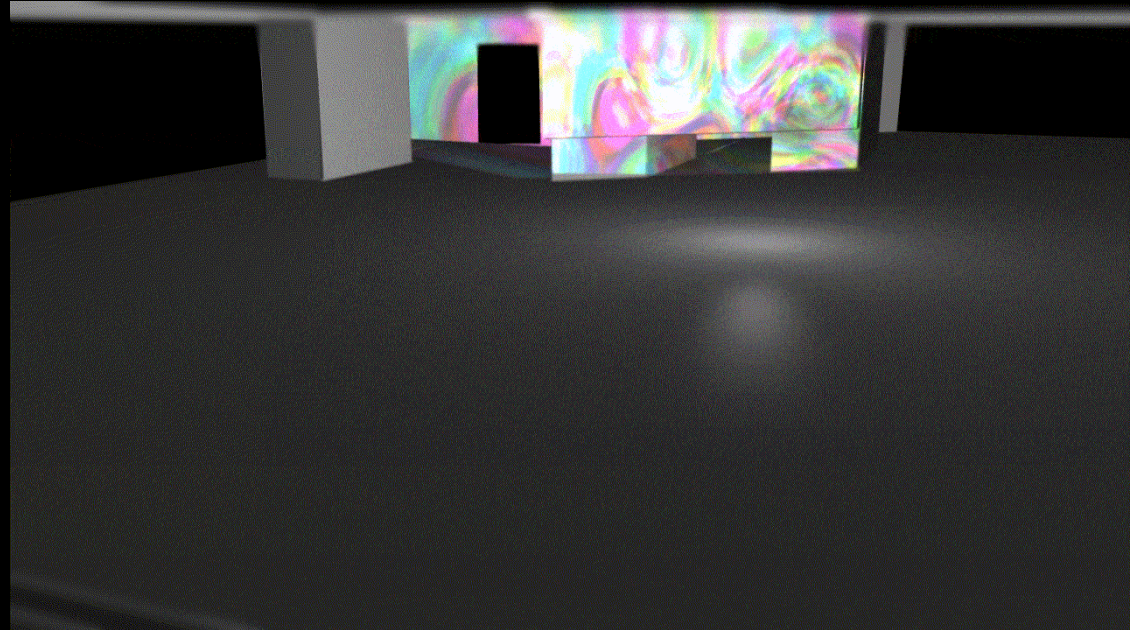
Geometry to instance

Poster format dimensions with
square starling



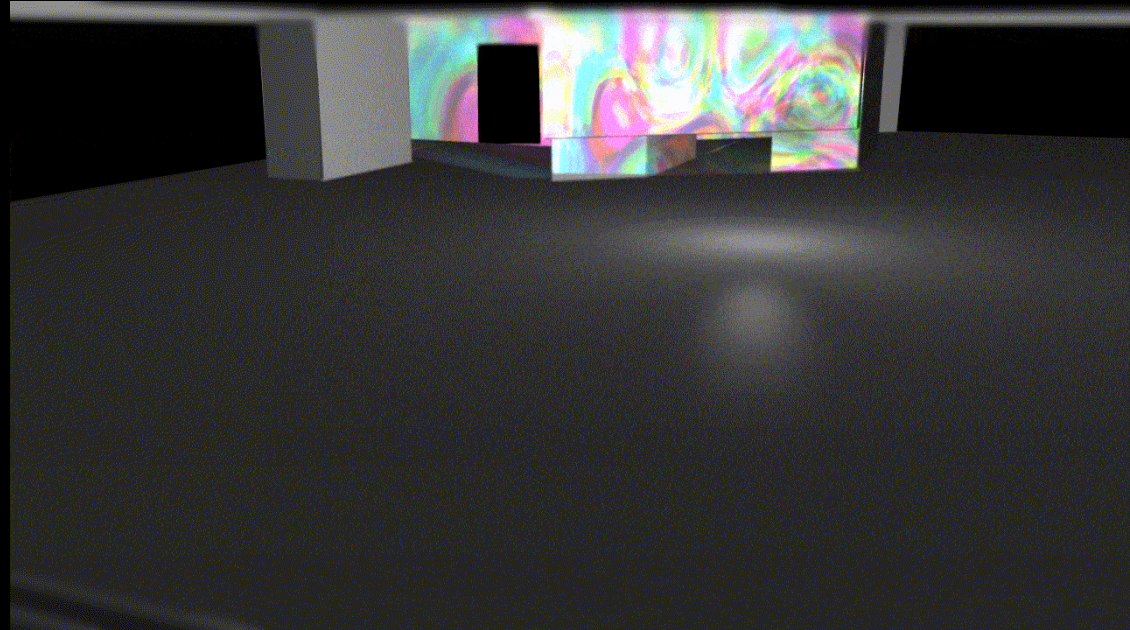
Starling Moire

CASE: MAIN
HALL SITUATION
RENDER

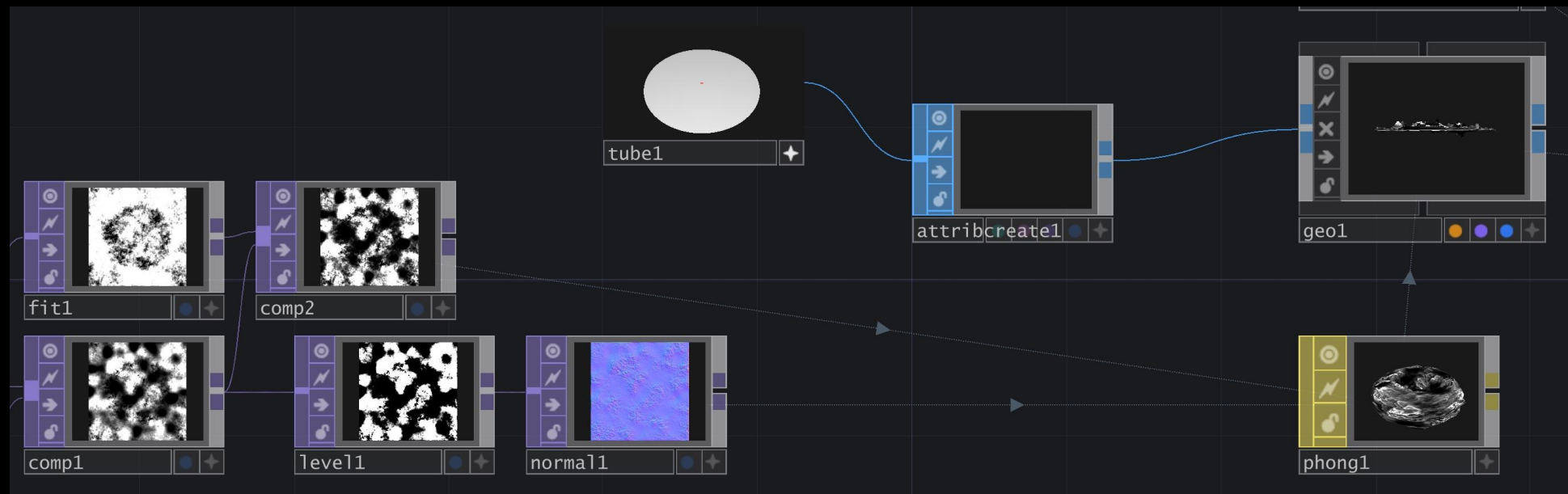


CASE: MAIN HALL SITUATION RENDER BREAKDOWN

- Main hall model
- Camera with human eye focal length and aperture
- Light for projection
- Light for ambient
- Geometry comp
- Render top
- Tilt shift effect tilted up



MATERIALS AND NORMAL MAPPING



MATERIALS AND NORMAL MAPPING

- So far we have just been working with blank geometry and projecting upon them. In the first class Operator oversight we also briefly brushed upon the fact that you can assign materials onto geometry.
- Material allows you to transform the blank geometry into something that looks more like a real life material. Or not, depending on the effect you're looking for.
- Touchdesigner also allows you to assign Normal Maps made up from images or videos which you can import or create generatively inside Touchdesigner.

DIFFERENCE MATERIAL AND NORMAL MAP

MATERIAL

JUST THE IMAGE OF THE TEXTURE YOU WANT THE GEOMETRY TO HAVE. IT HAS NO DEPTH INFORMATION, AND IT JUST WRAPS AROUND THE GEOMETRY

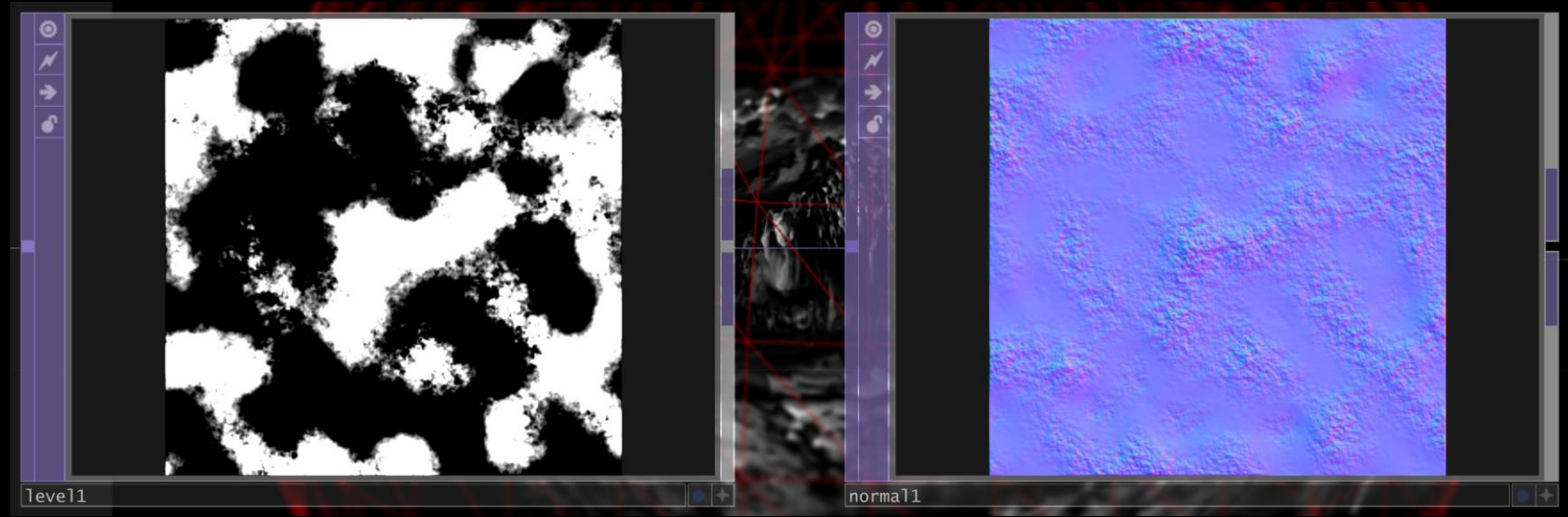


MATERIAL + NORMAL MAP

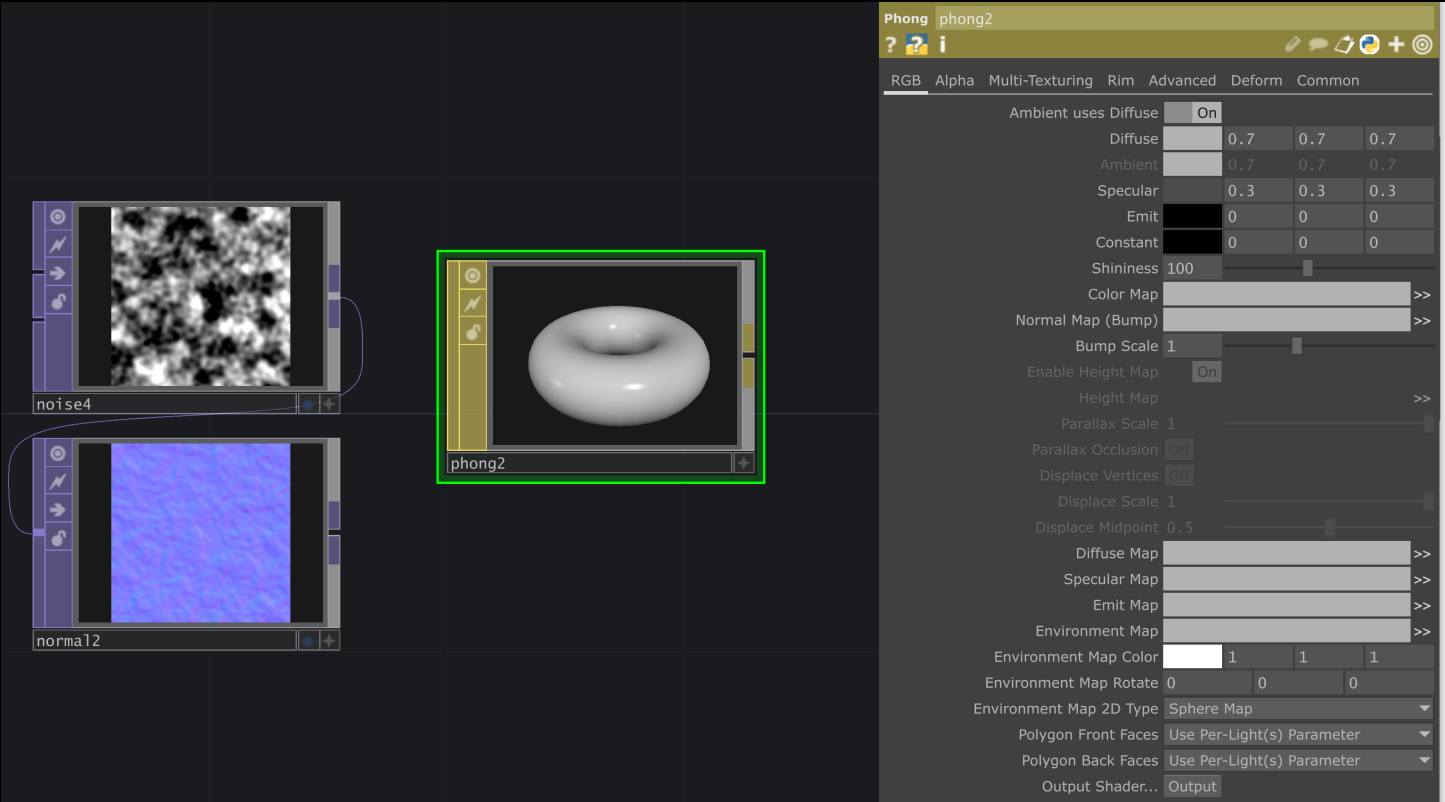
THE NORMAL MAP ADDS DEPTH INFORMATION TO THE MATERIAL GIVING THE ILLUSION OF DEPTH THROUGH LIGHTING AND REFLECTION. WHEN YOU LOOK CLOSELY, YOU'LL SEE THAT THE SURFACE IS STILL FLAT.

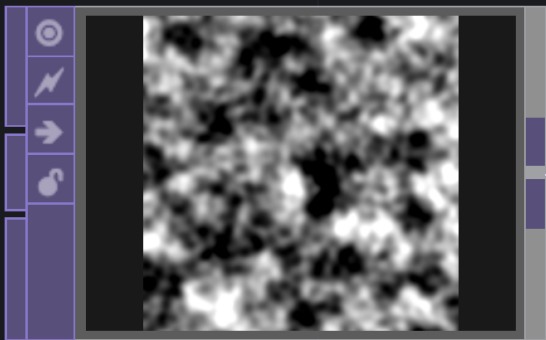


- A normal map in Touchdesigner is part of the TOP domain so you'll also find it there.
- The fact that it is part of the TOP domain allows you to easily derive normal maps from any TOP network you have going on.
- Normal maps are usually referenced into a *PHONG MAT*.

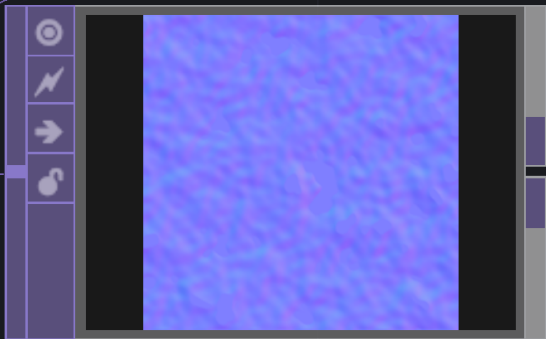


PHONG MATERIAL

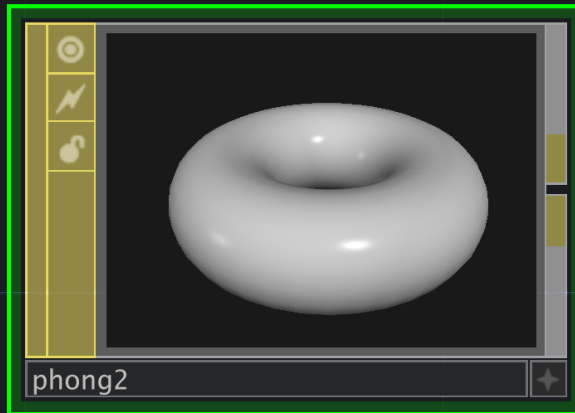




noise4



normal2



phong2

Phong phong2



RGB Alpha Multi-Texturing Rim Advanced Deform Common

Ambient uses Diffuse ☒ On

Diffuse

Ambient

Specular

Emit

Constant

Shininess

Color Map

Normal Map (Bump)

Bump Scale

Enable Height Map ☒ On

Height Map

Parallax Scale

Parallax Occlusion

Displace Vertices

Displace Scale

Displace Midpoint

Diffuse Map

Specular Map

Emit Map

Environment Map

Environment Map Color

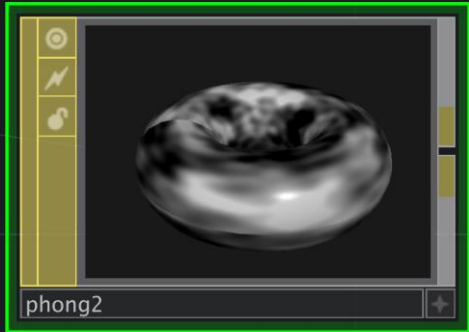
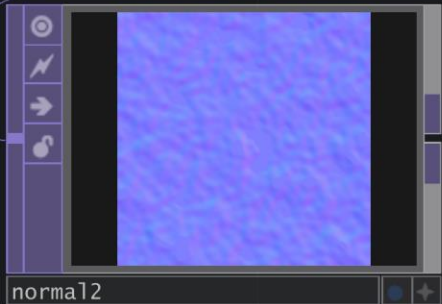
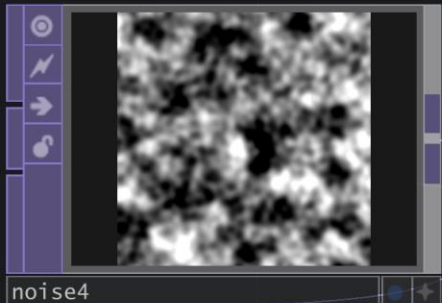
Environment Map Rotate

Environment Map 2D Type

Polygon Front Faces

Polygon Back Faces

Output Shader...



Phong phong2 ? ? i

RGB Alpha Multi-Texturing Rim Advanced Deform Common

Ambient uses Diffuse ☒ On

Diffuse 0.7 0.7 0.7

Ambient 0.7 0.7 0.7

Specular 0.3 0.3 0.3

Emit 0 0 0

Constant 0 0 0

Shininess 100

Color Map noise4 >>

Normal Map (Bump) >>

Bump Scale 1

Enable Height Map ☒ On

Height Map >>

Parallax Scale 1

Parallax Occlusion ☐ Off

Displace Vertices ☐ Off

Displace Scale 1

Displace Midpoint 0.5

Diffuse Map >>

Specular Map >>

Emit Map >>

Environment Map >>

Environment Map Color 1 1 1

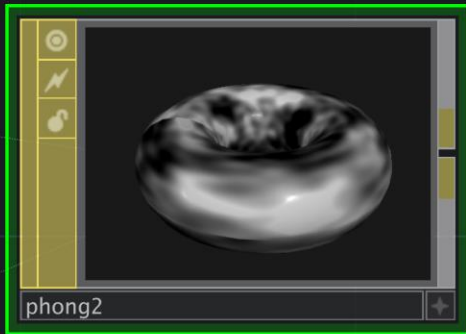
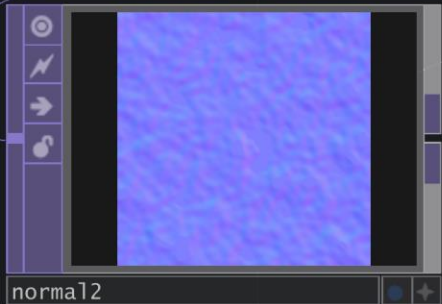
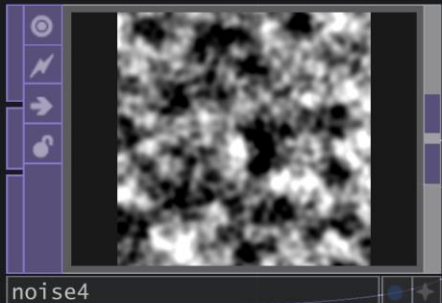
Environment Map Rotate 0 0 0

Environment Map 2D Type Sphere Map

Polygon Front Faces Use Per-Light(s) Parameter

Polygon Back Faces Use Per-Light(s) Parameter

Output Shader... Output



Phong phong2 ? ? i

RGB Alpha Multi-Texturing Rim Advanced Deform Common

Ambient uses Diffuse ☒ On

Diffuse 0.7 0.7 0.7

Ambient 0.7 0.7 0.7

Specular 0.3 0.3 0.3

Emit 0 0 0

Constant 0 0 0

Shininess 100

Color Map noise4 >>

Normal Map (Bump) normal2 >>

Bump Scale 1

Enable Height Map ☒ On

Height Map >>

Parallax Scale 1

Parallax Occlusion Off

Displace Vertices Off

Displace Scale 1

Displace Midpoint 0.5

Diffuse Map >>

Specular Map >>

Emit Map >>

Environment Map >>

Environment Map Color 1 1 1

Environment Map Rotate 0 0 0

Environment Map 2D Type Sphere Map

Polygon Front Faces Use Per-Light(s) Parameter

Polygon Back Faces Use Per-Light(s) Parameter

Output Shader... Output

Start: 1 End: 600 1 51 101 151 201 251 301 351 401 451 501 551 600

RStart: 1 REnd: 600

FPS: 60.0 Tempo: 120.0

ResetF: 1 T Sig: 4 4

TimeCode

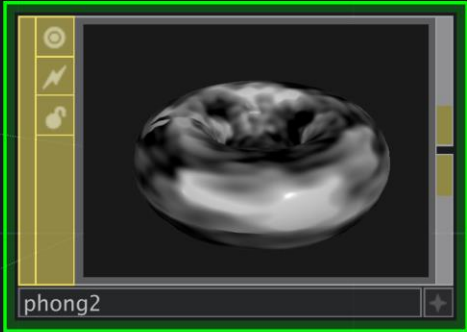
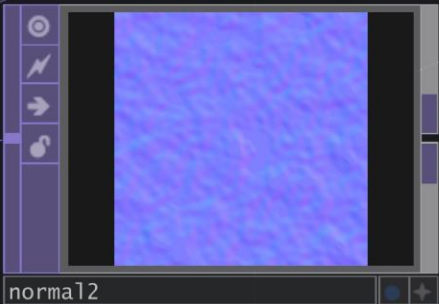
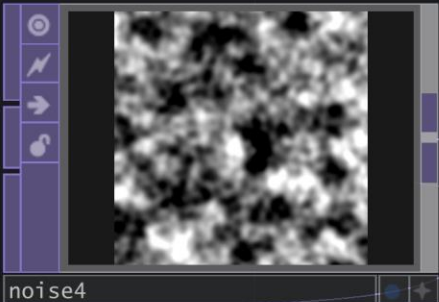
00:00:00:06

7

Range Limit

Loop

Once



Phong phong2 ? ? i

RGB Alpha Multi-Texturing Rim Advanced Deform Common

Ambient uses Diffuse ☒ On

Diffuse 0.7 0.7 0.7

Ambient 0.7 0.7 0.7

Specular 0.3 0.3 0.3

Emit 0 0 0

Constant 0 0 0

Shininess 100

Color Map noise4 >>

Normal Map (Bump) normal2 >>

Bump Scale 1

Enable Height Map ☒ On

Height Map noise4 >>

Parallax Scale 1

Parallax Occlusion Off

Displace Vertices Off

Displace Scale 1

Displace Midpoint 0.5

Diffuse Map >>

Specular Map >>

Emit Map >>

Environment Map >>

Environment Map Color 1 1 1

Environment Map Rotate 0 0 0

Environment Map 2D Type Sphere Map

Polygon Front Faces Use Per-Light(s) Parameter

Polygon Back Faces Use Per-Light(s) Parameter

Output Shader... Output

Start: 1 End: 600 1 51 101 151 201 251 301 351 401 451 501 551 600

RStart: 1 REnd: 600

FPS: 60.0 Tempo: 120.0

ResetF: 1 T Sig: 4 4

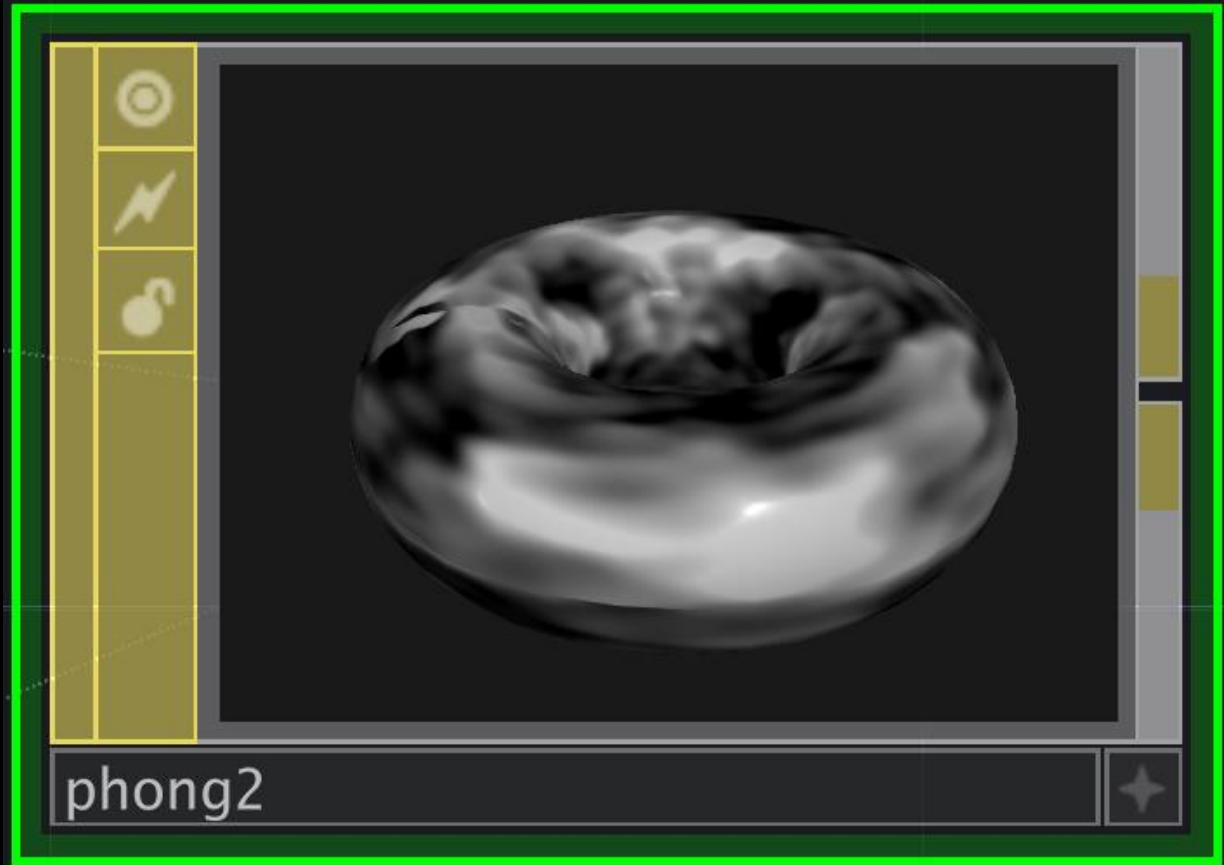
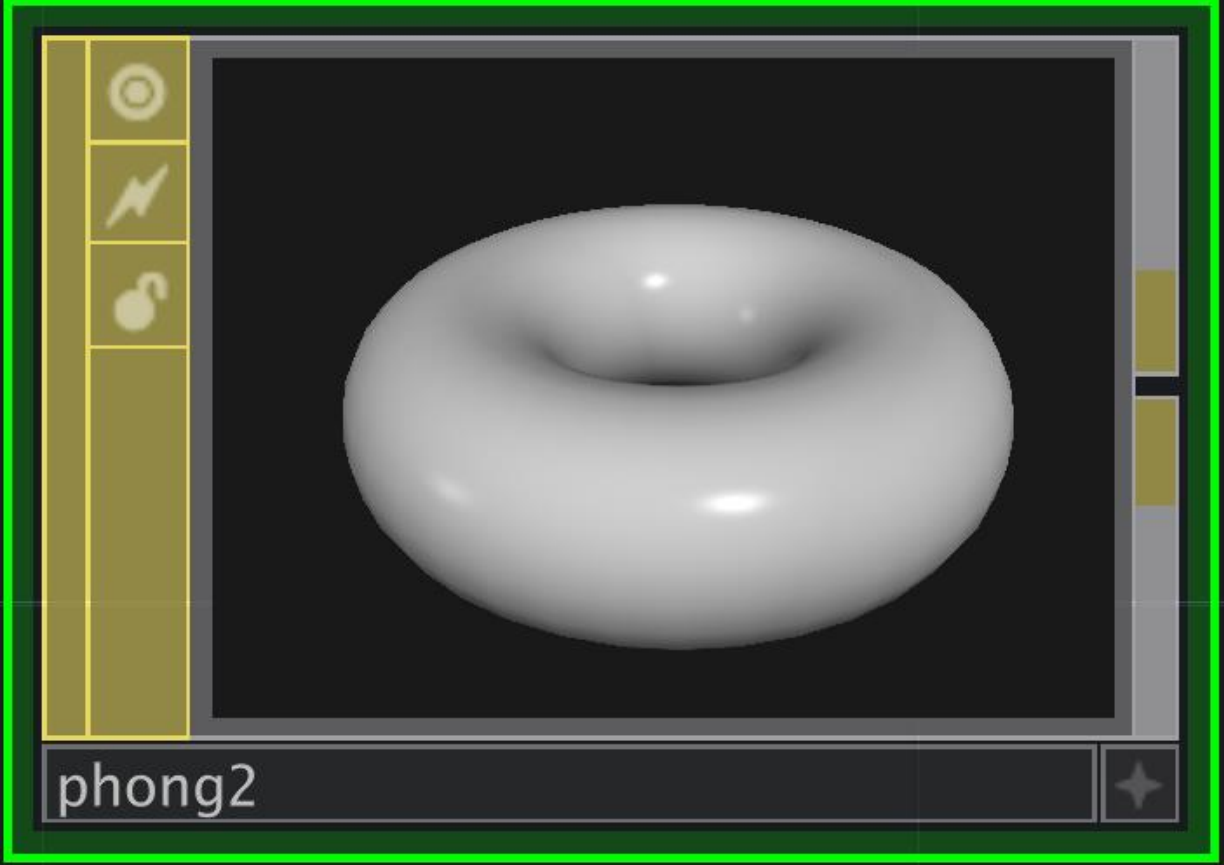
TimeCode

00:00:09:50

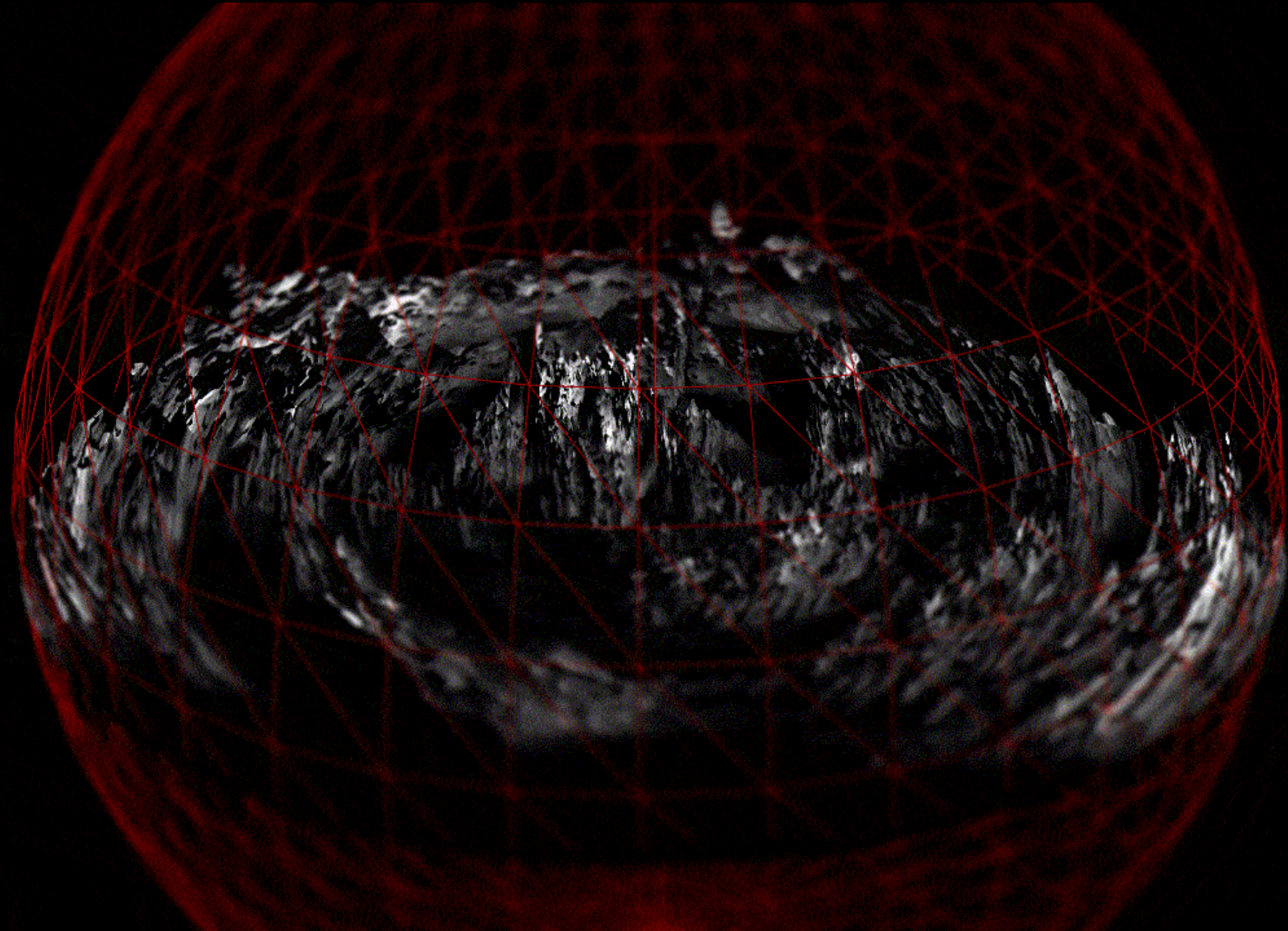
591

Range Limit

Loop Once



CASE: MAGIC SPHERE



- A normal mapping disc
- A source image for normal mapping
- Displacement of normal map
- Color map for normal map
- Wireframe material
- Sphere with wireframe material
- Displacement on the wireframe sphere
- Automated rotation of sphere and disc
- Camera
- Light
- Geometry
- Render
- Tilt shift
- Saving it as a movie/GIF

CASE: MAGIC SPHERE BREAKDOWN

